



DESCRIPTIVE STATISTICS PART 1

Session 38 — A complete guide to summarizing, measuring, and visualizing data distributions

Mean · Median · Mode

Variance · Std Dev

Percentiles · IQR

Box Plots · Histograms

Skewness · Kurtosis

Five-Number Summary

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Duration: 1h 52m · Instructor: Nitish Singh (CampusX)



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1 • What is Statistics? — The Big Picture

Statistics is the branch of mathematics that deals with the **collection, organization, analysis, interpretation, and presentation** of data. In the context of Machine Learning and Data Science, statistics provides the mathematical foundation for understanding data before building any model.

Why Statistics Matters for ML

Before you train any model, you **must** understand your data. Descriptive statistics helps you answer: What does my data look like? Where is the center? How spread out is it? Are there outliers? Is it symmetric? These answers drive critical decisions like feature selection, data cleaning, and model choice.

The Two Pillars of Statistics

Descriptive Statistics

Methods that **summarize and describe** the main features of a dataset.

- Works with the **data you have**
- Uses tables, charts, measures of center and spread
- No generalizations beyond the data
- **Examples:** Mean, Median, Histograms, Box Plots

Inferential Statistics

Methods that **draw conclusions about a population** from a sample.

- Makes **predictions and generalizations**
- Uses probability theory
- Involves uncertainty and confidence levels
- **Examples:** Hypothesis Testing, Confidence Intervals, Regression

2 • Population vs Sample

Population

The **entire group** of individuals, objects, or data points that you are interested in studying.

- Usually very large or infinite
- Often impractical or impossible to study completely
- Denoted by **N** (size)

Example: All 1.4 billion citizens of India (for a health survey).

Sample

A **representative subset** of the population selected for analysis.

- Smaller, manageable size
- Must be **randomly selected** to avoid bias
- Denoted by **n** (size)

Example: Surveying 10,000 randomly chosen citizens from India.

Sampling Bias

If the sample is **not representative** of the population, the conclusions drawn will be misleading. A biased sample leads to biased statistics. Always aim for **random sampling** where every member of the population has an equal chance of being selected.

Why Sample? Why Not Just Use the Entire Population?

- **Cost:** Studying every member is prohibitively expensive
- **Time:** Data collection from billions of data points takes too long
- **Practicality:** Some populations are infinite (e.g., all possible coin flips)
- **Destructive testing:** Testing might destroy the unit (e.g., crash-testing every car)

3 • Parameter vs Statistic

ASPECT	PARAMETER	STATISTIC
Describes	Entire Population	A Sample
Value	Fixed (usually unknown)	Varies from sample to sample
Notation (Mean)	μ (mu)	$\bar{x}$ (x-bar)
Notation (Std Dev)	σ (sigma)	s
Notation (Proportion)	p	$\hat{p}$ (p-hat)
Notation (Size)	N	n
Purpose	True characteristic of population	Estimate of the parameter

Example

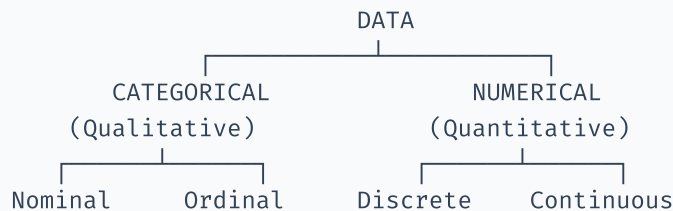
If the **average height of ALL adults in India** is 5'5" → this is a **parameter** ($\mu = 5'5"$).

If you measure 500 randomly selected adults and find their average height is 5'4.8" → this is a **statistic** ($\bar{x} = 5'4.8"$).

4 • Types of Data



Data Classification Hierarchy



4.1 Categorical (Qualitative) Data

Represents **characteristics, labels, or categories** — cannot be measured numerically in a meaningful way.

Nominal Data

- Categories with **no natural order**
- **Examples:** Gender (Male/Female), Blood Group (A/B/AB/O), City Names, Color
- Mathematical operations have no meaning

Ordinal Data

- Categories with a **natural ranking/order**
- The **differences between ranks are not equal**
- **Examples:** Education Level (High School < Bachelor's < Master's < PhD), Customer Rating (1–5 stars), T-Shirt Size (S < M < L < XL)

4.2 Numerical (Quantitative) Data

Represents **measurable quantities** — arithmetic operations are meaningful.

Discrete Data

- **Countable**, finite values — usually integers
- Cannot have fractional values between consecutive values
- **Examples:** Number of students (28), Number of cars sold (15), Dice rolls (1–6)

Continuous Data

- Can take **any value within a range** (infinitely many values)
- Measured, not counted

- **Examples:** Height (5.742 ft), Weight (68.3 kg), Temperature (37.2°C), Time (3.567 seconds)

ML Relevance

Understanding data types is critical because it determines: which **statistical measures** to apply (you can't compute mean of nominal data), which **visualizations** to use (bar chart vs histogram), and which **ML algorithms** are appropriate (classification vs regression). Most ML algorithms require **numerical input**, so categorical data must be encoded (One-Hot, Label Encoding).

5 • Descriptive vs Inferential Statistics

FEATURE	DESCRIPTIVE STATISTICS	INFERENTIAL STATISTICS
Goal	Summarize and describe data	Make predictions/generalizations about a population
Data Used	Entire dataset at hand	Sample data to infer about population
Uncertainty	No uncertainty — just describes what IS	Always involves uncertainty (confidence levels)
Tools	Mean, Median, Mode, Variance, Graphs	Hypothesis Tests, Confidence Intervals, p-values
Output	Tables, charts, summary numbers	Decisions, predictions, probability statements
Example	"The average marks in this class is 72"	"We are 95% confident the population mean is between 70–74"

6 • Types of Analysis

6.1 Univariate Analysis

Analysis of **a single variable** at a time. The goal is to describe the distribution, central tendency, and spread of one variable.

DATA TYPE	STATISTICAL MEASURES	VISUALIZATIONS
Categorical	Frequency, Mode, Proportion	Bar Chart, Pie Chart, Count Plot
Numerical	Mean, Median, Mode, Variance, Std Dev, Skewness, Kurtosis	Histogram, Box Plot, KDE Plot, Violin Plot

6.2 Bivariate Analysis

Analysis of **two variables** together to explore the relationship between them.

COMBINATION	TECHNIQUES	VISUALIZATIONS
Numerical–Numerical	Correlation (Pearson, Spearman)	Scatter Plot, Line Plot, Heatmap
Numerical–Categorical	Group means comparison, ANOVA	Box Plot, Violin Plot, Bar Plot (with error bars)
Categorical–Categorical	Chi-Square Test, Contingency Table	Stacked Bar Chart, Heatmap (crosstab)

6.3 Multivariate Analysis

Analysis of **three or more variables** simultaneously. Techniques include Pair Plots, PCA, Multiple Regression, and Parallel Coordinates.

7 • Measures of Central Tendency

Measures of central tendency identify a **single value** that best represents the "center" or typical value of a dataset.

7.1 Mean (Arithmetic Average)

The **sum of all values** divided by the **number of values**. The most commonly used measure of center.

FORMULA — MEAN

Population Mean: $\mu = (\sum x_i) / N$

Sample Mean: $\bar{x} = (\sum x_i) / n$

Example

Dataset: {4, 8, 6, 5, 3, 2, 8, 9, 2, 5}

Sum = 4+8+6+5+3+2+8+9+2+5 = 52

n = 10

Mean = 52 / 10 = 5.2

Properties of Mean

- Uses **every data point** in the calculation
- Highly **sensitive to outliers** (a single extreme value can shift it dramatically)
- Works best for **symmetric distributions**
- Suitable only for **numerical data** (interval and ratio scales)

7.2 Weighted Mean

When different values have different levels of **importance or frequency**, we assign weights to each value.

FORMULA — WEIGHTED MEAN

$$\bar{x}_w = \frac{\sum (w_i \cdot x_i)}{\sum (w_i)}$$

Example — CGPA Calculation

Subject grades: Math = 9 (4 credits), English = 7 (2 credits), Science = 8 (3 credits)

Weighted Mean = $(9 \times 4 + 7 \times 2 + 8 \times 3) / (4 + 2 + 3) = (36 + 14 + 24) / 9 = 74 / 9 \approx \mathbf{8.22}$

7.3 Trimmed Mean

To **reduce the effect of outliers**, a certain percentage of the smallest and largest values are removed before calculating the mean.

FORMULA — TRIMMED MEAN

1. Sort the data in ascending order
2. Remove the bottom p% and top p% of values
3. Calculate the mean of the remaining values

Example — 10% Trimmed Mean

Dataset (sorted): {1, 2, 3, 4, 5, 6, 7, 8, 9, 100}

Regular Mean = $145/10 = 14.5$ (pulled up by outlier 100)

10% trim → Remove 1 value from each end: {2, 3, 4, 5, 6, 7, 8, 9}

Trimmed Mean = $44/8 = 5.5$ (much more representative!)

7.4 Median

The **middle value** when data is arranged in ascending or descending order.

FORMULA — MEDIAN

If n is odd: Median = value at position $(n+1)/2$ If n is even: Median = average of values at positions $n/2$ and $(n/2)+1$

Properties of Median

- **Not affected by outliers** — robust measure
- Only depends on the **middle value(s)**
- Preferred over mean for **skewed distributions**
- Can be used for **ordinal data** as well

Example

Odd count: {3, 5, 7, 9, 11} → $n=5$ → Median = 7 (3rd position)

Even count: {3, 5, 7, 9} → $n=4$ → Median = $(5+7)/2 = 6$

With outlier: {3, 5, 7, 9, 1000} → Mean = 204.8, but Median = **7** (unaffected!)

7.5 Mode

The value that **appears most frequently** in the dataset.

Properties of Mode

- The **only** measure of central tendency applicable to **nominal (categorical) data**
- A dataset can be **unimodal** (1 mode), **bimodal** (2 modes), **multimodal** (3+ modes), or have **no mode**
- Not affected by outliers
- Useful for finding the most popular or common category

7.6 When to Use What?

MEASURE	BEST FOR	AVOID WHEN
Mean	Symmetric distributions, no outliers	Skewed data, extreme outliers present
Median	Skewed distributions, outliers present, ordinal data	Small datasets where every value matters
Mode	Categorical data, finding most common value	Continuous data (usually no repeats)
Trimmed Mean	Data with suspected outliers, wanting robustness	Small datasets (too much data lost)
Weighted Mean	Values have different importance/frequency	All values equally important

8 • Measures of Dispersion (Spread)

Central tendency alone is not enough. Two datasets can have the **same mean** but very **different distributions**. Dispersion measures tell us **how spread out** the data is.

Why Dispersion Matters

Dataset A: {49, 50, 51} → Mean = 50

Dataset B: {0, 50, 100} → Mean = 50

Same mean, but A's values are tightly clustered while B's are wildly spread. Dispersion captures this difference!

8.1 Range

The simplest measure of spread — the **difference between the maximum and minimum** values.

FORMULA — RANGE

Range = Maximum Value - Minimum Value

- **Pros:** Simple and intuitive
- **Cons:** Uses only 2 data points; highly sensitive to outliers; tells nothing about how data is distributed between min and max

8.2 Variance

Measures the **average of squared deviations** from the mean. Tells us how far each data point is from the center, on average.

FORMULA — VARIANCE

Population Variance: $\sigma^2 = \sum (x_i - \mu)^2 / N$

Sample Variance: $s^2 = \sum (x_i - \bar{x})^2 / (n - 1)$

🔑 Why divide by (n-1) for sample variance?

This is called **Bessel's Correction**. When we calculate variance from a sample, we tend to **underestimate** the true population variance because the sample mean is closer to the sample data than the true population mean. Dividing by (n-1) instead of n corrects this bias, giving us an **unbiased estimator** of population variance. This concept is called **degrees of freedom** — once we know the sample mean and n-1 values, the last value is determined.

Step-by-step Variance Calculation

📝 Example

Dataset: {2, 4, 4, 4, 5, 5, 7, 9}

Step 1: Calculate the mean: $\bar{x} = (2+4+4+4+5+5+7+9)/8 = 40/8 = 5$

Step 2: Subtract the mean from each value (deviations):

$(2-5)=-3, (4-5)=-1, (4-5)=-1, (4-5)=-1, (5-5)=0, (5-5)=0, (7-5)=2, (9-5)=4$

Step 3: Square each deviation:

9, 1, 1, 1, 0, 0, 4, 16

Step 4: Sum of squared deviations = 32

Step 5: Population variance $\sigma^2 = 32/8 = 4$

Sample variance $s^2 = 32/7 \approx 4.57$

8.3 Standard Deviation

The **square root of variance**. It brings the measure back to the **same units** as the original data, making it more interpretable.

FORMULA — STANDARD DEVIATION

Population: $\sigma = \sqrt{(\sigma^2)} = \sqrt{[\sum(x_i - \mu)^2 / N]}$

Sample: $s = \sqrt{(s^2)} = \sqrt{[\sum(x_i - \bar{x})^2 / (n - 1)]}$

Interpreting Standard Deviation

- **Small σ** → data points are **clustered close** to the mean
- **Large σ** → data points are **spread far** from the mean
- For a **normal distribution** (Empirical Rule / 68–95–99.7 Rule):
 - ~68% of data falls within $\mu \pm 1\sigma$
 - ~95% of data falls within $\mu \pm 2\sigma$

- ~99.7% of data falls within $\mu \pm 3\sigma$

8.4 Coefficient of Variation (CV)

A **relative measure** of variability that normalizes standard deviation by the mean. Used to **compare variability** across datasets with different units or scales.

FORMULA — COEFFICIENT OF VARIATION

$$CV = (\text{Standard Deviation} / \text{Mean}) \times 100\%$$

$$CV = (\sigma / \mu) \times 100\%$$

Example — Comparing Variability

Heights: Mean = 170 cm, SD = 10 cm → $CV = (10/170) \times 100 = 5.88\%$

Weights: Mean = 70 kg, SD = 8 kg → $CV = (8/70) \times 100 = 11.43\%$

Even though SD of heights (10) > SD of weights (8), weights have **higher relative variability** (11.43% > 5.88%).

9 • Percentiles, Quartiles & IQR

9.1 Percentiles

A **percentile** is a value below which a given percentage of data falls. The k th percentile (P_k) means that $k\%$ of the data values are less than or equal to that value.

FORMULA — PERCENTILE POSITION

Position = $(k / 100) \times (n + 1)$ where k = desired percentile, n = number of data points

Example

If your exam score is at the **90th percentile**, it means **90% of students scored lower** than you. Only 10% scored higher.

9.2 Quartiles

Quartiles divide the ordered dataset into **four equal parts**.

QUARTILE	PERCENTILE	MEANING
Q1 (First Quartile)	25th Percentile	25% of data falls below this value
Q2 (Second Quartile)	50th Percentile	The Median — 50% of data below
Q3 (Third Quartile)	75th Percentile	75% of data falls below this value

9.3 Interquartile Range (IQR)

The range of the **middle 50%** of the data — the distance between Q3 and Q1.

FORMULA — IQR

$$IQR = Q3 - Q1$$

Why IQR is Important

- **Robust:** Not affected by outliers (unlike range)
- **Used for outlier detection:** Any value below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ is a potential outlier
- Forms the basis of **box plots**

 Example — Complete Calculation

Dataset (sorted): {3, 5, 7, 8, 12, 13, 14, 18, 21}

$n = 9$

Q1 = Median of lower half {3, 5, 7, 8} = $(5+7)/2 = 6$

Q2 (Median) = **12** (5th value)

Q3 = Median of upper half {13, 14, 18, 21} = $(14+18)/2 = 16$

IQR = $16 - 6 = 10$

Outlier Fences:

Lower Fence = $Q1 - 1.5 \times IQR = 6 - 15 = -9$

Upper Fence = $Q3 + 1.5 \times IQR = 16 + 15 = 31$

Any value < -9 or > 31 would be an outlier. All values are within bounds ✓

10 • Five-Number Summary

A concise summary that captures the key features of a data distribution using just **five values**.



The Five-Number Summary

Min	Q1 (25%)	Median (50%)	Q3 (75%)	Max
-----	-------------	-----------------	-------------	-----

COMPONENT	WHAT IT TELLS YOU
Minimum	The smallest value in the dataset
Q1 (25th Percentile)	Below this: 25% of data; boundary of the lower quarter
Median (Q2)	The center point — 50% above, 50% below
Q3 (75th Percentile)	Below this: 75% of data; boundary of the upper quarter
Maximum	The largest value in the dataset



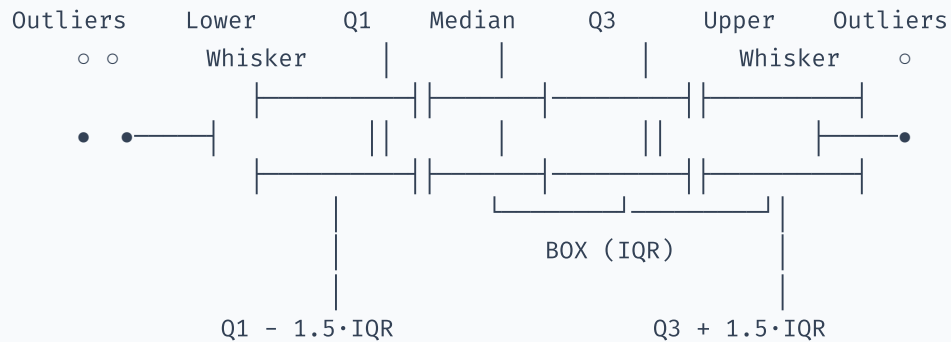
What You Can Derive

From the five-number summary, you can immediately calculate: **Range** ($\text{Max} - \text{Min}$), **IQR** ($\text{Q3} - \text{Q1}$), **detect skewness** (if Median is closer to Q1 → right-skewed; closer to Q3 → left-skewed), and **identify outliers** using the $1.5 \times \text{IQR}$ rule.

11 · Box Plot (Box-and-Whisker Plot)

A **visual representation** of the five-number summary that also highlights outliers. One of the most powerful tools for univariate analysis.

Anatomy of a Box Plot



Components Explained

- **Box:** Spans from Q1 to Q3 (contains the middle 50% of data = IQR)
- **Median line:** A line inside the box showing Q2
- **Whiskers:** Extend from the box to the farthest data point **within** $1.5 \times IQR$ from Q1 and Q3
- **Outliers:** Individual points plotted beyond the whiskers (beyond $\pm 1.5 \times IQR$ from Q1/Q3)

Reading a Box Plot

OBSERVATION	INTERPRETATION
Median line near center of box	Data is approximately symmetric
Median line closer to Q1	Data is right-skewed (positive skew)
Median line closer to Q3	Data is left-skewed (negative skew)
Long whiskers	Data has a wide spread
Many outlier dots	Distribution has heavy tails
Short box (small IQR)	Data is tightly clustered around the median

 ML Use Case — Outlier Detection

Box plots are the go-to visualization for **spotting outliers** during EDA (Exploratory Data Analysis). In ML pipelines, outliers can drastically affect model training. Use box plots on each feature to decide whether to remove, cap, or transform outliers.

12 · Frequency Distribution & Histograms

12.1 Frequency Distribution Table

Organizes data by showing **how many times** each value (or range of values) appears.

CLASS INTERVAL (BINS)	FREQUENCY (F)	RELATIVE FREQUENCY	CUMULATIVE FREQUENCY
0 – 20	5	$5/30 = 0.167$	5
20 – 40	8	$8/30 = 0.267$	13
40 – 60	10	$10/30 = 0.333$	23
60 – 80	5	$5/30 = 0.167$	28
80 – 100	2	$2/30 = 0.067$	30

Key Terms

- **Frequency (f):** Count of values in each bin
- **Relative Frequency:** Proportion = $f / \text{total } n$ (sums to 1)
- **Cumulative Frequency:** Running total of frequencies
- **Bin Width:** The range of values in each interval

12.2 Histogram

A **graphical representation** of a frequency distribution for numerical data. Unlike bar charts, histogram bars **touch each other** (continuous data, no gaps).

Histogram vs Bar Chart

FEATURE	HISTOGRAM	BAR CHART
Data Type	Numerical (continuous)	Categorical
Bars	Touch each other (no gaps)	Have gaps between them
X-axis	Continuous intervals (bins)	Discrete categories
Y-axis	Frequency / Density	Frequency / Count
Bar order	Cannot be rearranged	Can be rearranged

 Choosing the Number of Bins

Sturges' Rule: $k = 1 + 3.322 \times \log_{10}(n)$

Square Root Rule: $k = \sqrt{n}$

Too few bins → oversimplified, hides patterns. Too many bins → too noisy, hard to interpret. Python's `matplotlib` and `seaborn` handle this automatically with `bins='auto'` .

13 · Skewness — Measuring Asymmetry

Skewness measures the **degree and direction of asymmetry** in a distribution. A perfectly symmetric distribution (like a perfect bell curve) has skewness = 0.

Types of Skewness

LEFT-SKEWED
(Negative Skew)



Tail ← LEFT

Mode > Median > Mean

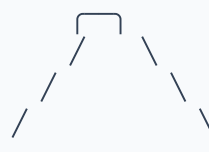
SYMMETRIC
(No Skew)



Mean=Median=Mode

All Equal

RIGHT-SKEWED
(Positive Skew)



Tail → RIGHT

Mean > Median > Mode

Understanding Skewness Values

SKEWNESS VALUE	INTERPRETATION	TAIL DIRECTION	RELATIONSHIP
= 0	Perfectly Symmetric	Equal tails	Mean = Median = Mode
> 0 (Positive)	Right-skewed	Longer right tail	Mean > Median > Mode
< 0 (Negative)	Left-skewed	Longer left tail	Mode > Median > Mean
skew < 0.5	Approximately Symmetric	—	—
0.5 < skew < 1	Moderately Skewed	—	—
skew > 1	Highly Skewed	—	—

FORMULA — PEARSON'S SKEWNESS COEFFICIENT

Skewness = $3 \times (\text{Mean} - \text{Median}) / \text{Standard Deviation}$

 Real-World Examples

Right-skewed (Positive): Income distribution (most people earn moderate amounts, a few earn extremely high), house prices, insurance claims.

Left-skewed (Negative): Age at retirement (most retire around 60–65, few retire very young), exam scores in an easy test (most score high).

Symmetric: Heights and weights of a population, measurement errors.

 ML Implications

Many ML algorithms assume **normally distributed** (symmetric) features. Highly skewed features can lead to poor model performance. Common fixes: **Log transformation** (for right-skewed), **Square root transformation**, **Box-Cox transformation**, or **binning**.

14 · Kurtosis — Measuring Tail Heaviness

Kurtosis measures the **"tailedness"** of a distribution — how much data is in the tails vs the center. It tells you about the **frequency of extreme values (outliers)**.

⚠ Common Misconception

Kurtosis is **NOT** about the **"peakedness"** of a distribution. It's about the **tails**. A high kurtosis means more data in the tails (more extreme outliers), not necessarily a taller peak.

Types of Kurtosis

TYPE	KURTOSIS VALUE	EXCESS KURTOSIS	CHARACTERISTICS	OUTLIERS
Mesokurtic	= 3	= 0	Normal distribution (baseline)	Normal frequency
Leptokurtic	> 3	> 0	Heavier tails, sharper peak	More extreme outliers
Platykurtic	< 3	< 0	Lighter tails, flatter peak	Fewer extreme outliers

FORMULA — EXCESS KURTOSIS

Excess Kurtosis = Kurtosis - 3 (Normal distribution has Excess Kurtosis = 0)

📝 Real-World Examples

Leptokurtic (heavy tails): Stock market returns (frequent extreme gains/losses), income distribution.

Platykurtic (light tails): Uniform distribution, well-controlled manufacturing processes.

Mesokurtic (normal tails): Height distribution, measurement errors.

💡 Why It Matters in ML

High kurtosis → your data has **more outliers** → models like Linear Regression will be heavily affected. Consider **robust models** (Random Forest, Gradient Boosting) or **outlier treatment** before training. Also relevant in **financial modeling** where fat tails (leptokurtic) mean higher risk.

15 · Python Implementation

Here's how to compute all the descriptive statistics covered in this session using Python:

15.1 Basic Measures with NumPy & Pandas

```
import numpy as np
import pandas as pd
from scipy import stats

# Sample data
data = [23, 45, 12, 67, 34, 89, 23, 56, 78, 12, 45, 67, 34, 91, 15]
df = pd.Series(data)

# — Measures of Central Tendency —
print("Mean:      ", np.mean(data))          # or df.mean()
print("Median:    ", np.median(data))        # or df.median()
print("Mode:      ", stats.mode(data).mode)   # scipy mode
print("Trimmed Mean: ", stats.trim_mean(data, 0.1)) # 10% trim

# — Measures of Dispersion —
print("Range:     ", np.ptp(data))            # peak-to-peak
print("Variance:  ", np.var(data, ddof=1))    # sample var
print("Std Dev:   ", np.std(data, ddof=1))    # sample SD
print("CV:        ", (np.std(data, ddof=1) / np.mean(data)) * 100)

# — Percentiles & Quartiles —
print("Q1:        ", np.percentile(data, 25))
print("Q2 (Median): ", np.percentile(data, 50))
print("Q3:        ", np.percentile(data, 75))
print("IQR:       ", np.percentile(data, 75) - np.percentile(data, 25))

# — Five-Number Summary (at once!) —
print(df.describe()) # gives count, mean, std, min, 25%, 50%, 75%, max

# — Shape Measures —
print("Skewness:   ", df.skew())              # or stats.skew(data)
print("Kurtosis:   ", df.kurtosis())          # excess kurtosis
```

15.2 Visualization

```
import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(1, 3, figsize=(15, 4))

# Histogram
axes[0].hist(data, bins='auto', color='#3b82f6', edgecolor='white', alpha=0.8)
axes[0].set_title('Histogram')
axes[0].axvline(np.mean(data), color='red', linestyle='--', label='Mean')
axes[0].axvline(np.median(data), color='green', linestyle='--', label='Median')
axes[0].legend()

# Box Plot
axes[1].boxplot(data, vert=True)
axes[1].set_title('Box Plot')

# KDE Plot (smooth histogram)
sns.kdeplot(data, ax=axes[2], fill=True, color='#8b5cf6')
axes[2].set_title('KDE Plot (Density)')

plt.tight_layout()
plt.show()
```

15.3 Outlier Detection with IQR

```
Q1 = np.percentile(data, 25)
Q3 = np.percentile(data, 75)
IQR = Q3 - Q1

lower_fence = Q1 - 1.5 * IQR
upper_fence = Q3 + 1.5 * IQR

outliers = [x for x in data if x < lower_fence or x > upper_fence]
print(f"Lower Fence: {lower_fence}")
print(f"Upper Fence: {upper_fence}")
print(f"Outliers: {outliers}")
```

16 · Quick Reference Card

Descriptive Statistics — Complete Cheat Sheet

Measures of Central Tendency

- **Mean** = $\sum x_i / n$ — sensitive to outliers
- **Weighted Mean** = $\sum (w_i \cdot x_i) / \sum w_i$ — accounts for importance
- **Trimmed Mean** — remove p% from both ends, then calculate mean
- **Median** = middle value — robust to outliers
- **Mode** = most frequent value — works on categorical data

Measures of Dispersion

- **Range** = Max – Min
- **Variance** = $\sum (x_i - \bar{x})^2 / (n-1)$ — squared units
- **Standard Deviation** = $\sqrt{\text{Variance}}$ — same units as data
- **CV** = $(\text{SD} / \text{Mean}) \times 100\%$ — relative variability
- **IQR** = $Q3 - Q1$ — spread of middle 50%

Shape Measures

- **Skewness** = 0 (symmetric), >0 (right-skewed), <0 (left-skewed)
- **Kurtosis** = 3 (normal), >3 (heavy tails), <3 (light tails)

Outlier Detection (IQR Method)

- Lower Fence = $Q1 - 1.5 \times \text{IQR}$
- Upper Fence = $Q3 + 1.5 \times \text{IQR}$
- Any point outside fences → potential outlier

Python Quick Reference

- `df.describe()` — Five-number summary + mean, std, count
- `df.skew()` — Skewness
- `df.kurtosis()` — Excess Kurtosis
- `scipy.stats.trim_mean(data, 0.1)` — 10% trimmed mean
- `sns.boxplot(data=df)` — Boxplot
- `plt.hist(data, bins='auto')` — Histogram

 What's Next? → Descriptive Statistics Part 2 (Session 39)

Part 2 continues with advanced descriptive statistics topics including more on data visualization techniques, bivariate analysis in depth, correlation analysis, and practical applications with real-world datasets using Python.